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# Computational modeling of UHF/SHF electromagnetic field propagation and intensity in enclosures with dielectric materials

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## Abstract

The study of the spatial distribution of indoor electromagnetic waves is a fundamental problem in applied electromagnetism due to attenuation, dielectric absorption, and shielding phenomena caused by building structures. In this work, a two-dimensional computational model is developed to simulate stationary propagation and electromagnetic field intensity in the UHF ($2.4\ \mathrm{GHz}$, $\lambda = 12.5\ \mathrm{cm}$) and SHF ($5.0\ \mathrm{GHz}$, $\lambda = 6.0\ \mathrm{cm}$) bands within enclosures containing dielectric and conductive materials. The spatial domain is discretized using the centered finite-difference method, applying Dirichlet boundary conditions at radiating sources and absorbing boundaries, while incorporating local loss coefficients based on the relative permittivity ($\varepsilon_r$) and conductivity ($\sigma$) of each material (drywall, brick, concrete, and metal). The resulting algebraic system is iteratively solved via Successive Over-Relaxation (SOR), determining at each node the received power ($\mathrm{dBm}$), the time-averaged Poynting vector magnitude ($\mu\mathrm{W/m^2}$), and the root-mean-square electric field $E_{\mathrm{rms}}$ ($\mathrm{V/m}$). Validation against the analytical log-distance and ITU-R P.1238 models yields a high correlation ($R^2 = 0.978$) and a root-mean-square error of $1.84\ \mathrm{dB}$, confirming the higher dielectric penetration of the $2.4\ \mathrm{GHz}$ band and the Faraday cage shielding effect in metallic enclosures.

**Keywords:** *Electromagnetic Waves; Dielectric Attenuation; Poynting Vector; Finite Differences; Successive Over-Relaxation (SOR).* | **Modality:** *Computational.*

## Introduction and motivation

When a microwave electromagnetic wave propagates inside an enclosed space, its energy flux density decreases due to both geometric wavefront divergence and dissipation within walls characterized by relative permittivity $\varepsilon_r$ and effective conductivity $\sigma > 0$ [1, 2]. While closed-form free-space analytical equations are only valid in homogeneous media, numerical modeling enables solving realistic architectural geometries.

**General objective:** To numerically model and validate the spatial distribution of received power ($P_{\mathrm{rx}}$), time-averaged Poynting vector magnitude ($\langle S \rangle$), and root-mean-square electric field intensity ($E_{\mathrm{rms}}$) in the $2.4\ \mathrm{GHz}$ and $5.0\ \mathrm{GHz}$ bands inside 2D enclosures with dielectric and conductive materials using finite differences and SOR relaxation.
**Specific objectives:** (1) Discretize the 2D domain via finite differences and solve with SOR. (2) Quantify dielectric absorption across $\varepsilon_r$ and $\sigma$. (3) Model Faraday cage shielding and validate against ITU-R P.1238.

## Theoretical framework / Physical model

In air (intrinsic impedance $\eta_0 = \sqrt{\mu_0/\varepsilon_0} \approx 376.7\,\Omega$), the time-averaged magnitude of the **Poynting vector** $\langle S \rangle$ ($\mathrm{W/m^2}$) relates to the RMS electric field $E_{\mathrm{rms}}$ ($\mathrm{V/m}$), the RMS magnetic field $H_{\mathrm{rms}}$ ($\mathrm{A/m}$), and the received power $P_{\mathrm{rx}}$ captured by an isotropic effective aperture $A_{\mathrm{eff}} = \lambda^2/(4\pi)$ as follows [1]:

$$\langle S \rangle = \frac{E_{\mathrm{rms}}^2}{\eta_0} = \frac{4\pi\, P_{\mathrm{rx}}}{\lambda^2} \implies E_{\mathrm{rms}} = \sqrt{\frac{4\pi\,\eta_0\, P_{\mathrm{rx}}}{\lambda^2}}, \quad H_{\mathrm{rms}} = \frac{E_{\mathrm{rms}}}{\eta_0} \qquad (1)$$

When traversing a lossy dielectric material or a conductor, the field attenuates exponentially according to the Beer-Lambert law $E(x) = E_0 e^{-\alpha x}$. In a 2D stationary regime, the normalized energy intensity envelope $u(x, y) \in [0, 100]$ over the domain $\Omega$ with spatial absorption $\kappa^2(x, y)$ is governed by the elliptic partial differential equation [3]:

$$\nabla^2 u(x, y) - \kappa^2(x, y)\, u(x, y) = 0 \quad \text{in } \Omega \qquad (2)$$

subject to fixed **Dirichlet** boundary conditions at the radiating source $u(x_0, y_0) = u_{\mathrm{EIRP}}$ and null absorbing boundaries along the outer perimeter $u|_{\partial\Omega} = 0$.

> **Physical variable nomenclature:**
> • $\langle S \rangle$: Time-averaged Poynting vector ($\mu\mathrm{W/m^2}$, net energy flux per unit area).
> • $E_{\mathrm{rms}}$, $H_{\mathrm{rms}}$: Root-mean-square electric ($\mathrm{V/m}$) and magnetic ($\mathrm{A/m}$) field strengths.
> • $\eta_0$: Free-space intrinsic wave impedance ($\sqrt{\mu_0/\varepsilon_0} \approx 376.73\,\Omega \approx 120\pi\,\Omega$).
> • $P_{\mathrm{rx}}$: Received power at isotropic antenna ($A_{\mathrm{eff}} = \lambda^2/4\pi$) in dBm converted to Watts.
> • $u(x, y)$: Normalized energy intensity envelope ($0 \le u \le 100$); $\kappa(x, y)$: Spatial dissipation ($m^{-1}$).

## Description of the modeled system

The developed software (*SimuNet Física*) discretizes $12\ \mathrm{m} \times 12\ \mathrm{m}$ enclosures into $24 \times 24$ node grids ($h = 0.5\ \mathrm{m}$), allowing placement of sources operating at $2.4\ \mathrm{GHz}$ ($\lambda = 12.5\ \mathrm{cm}$) and $5.0\ \mathrm{GHz}$ ($\lambda = 6.0\ \mathrm{cm}$) as well as walls with the constitutive properties listed in Table 1 [4].

**Table 1.** Electromagnetic properties and attenuation losses of modeled media (ITU-R P.2040).

| Material / Medium | Permittivity $\varepsilon_r$ | Conductivity $\sigma$ (S/m) | Loss (dB) | Base Coeff. $\alpha_{i,j}$ |
|---|---|---|---|---|
| **Air (Free Space)** | 1.0006 | 0.000 | 0 dB | 0.0012 |
| **Drywall (Plasterboard)** | 2.8 | 0.015 | 3 dB | 0.0200 |
| **Brick Wall** | 4.5 | 0.038 | 8 dB | 0.0800 |
| **Reinforced Concrete** | 6.5 | 0.085 | 15 dB | 0.1500 |
| **Metal (Faraday Cage)** | — | $3.5 \times 10^7$ | $\geq 35$ dB | 0.2500 |

**Base Coefficient ($\alpha_{i,j}$):** Dimensionless nodal dielectric attenuation factor at $2.4\ \mathrm{GHz}$ (ITU-R P.2040) derived from medium permittivity and conductivity. At $5.0\ \mathrm{GHz}$, the factor scales by $1.4\times$ due to increased dispersion and molecular absorption.

## Computational methodology

- **5-Point Finite-Difference Discretization:** Approximando the Laplacian operator on the 2D grid with the local dielectric absorption factor $\alpha_{i,j}(f, \varepsilon_r, \sigma)$:

$$u_{i,j}^{\mathrm{GS}} = (1 - \alpha_{i,j})\, \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}}{4} \qquad (3)$$

- **Successive Over-Relaxation (SOR) Solver:** Extrapolating the Gauss-Seidel step ($\omega = 1$) using the optimal relaxation factor $\omega \in (1, 2)$ until satisfying $\|u^{(k+1)} - u^{(k)}\|_\infty < \varepsilon$ [3]:

$$u_{i,j}^{(k+1)} = (1 - \omega)\, u_{i,j}^{(k)} + \omega\, u_{i,j}^{\mathrm{GS}} \qquad (4)$$

  where $\omega_{\mathrm{opt}} = 1.45$ minimizes the spectral radius of iterative convergence with tolerance $\varepsilon = 10^{-4}$.
- **Analytical Validation:** Contrasting the numerical radial profile against the log-distance and ITU-R P.1238 indoor model: $P_{\mathrm{rx}}(d) = P_0(d_0) - 10n \log_{10}(d/d_0) - \sum L_{\mathrm{wall}}$ [5].

## Results and discussion

Figure 1 presents the two-dimensional distribution of received power $P_{\mathrm{rx}}$ and electric field obtained numerically. It is verified that the UHF band ($2.4\ \mathrm{GHz}$, $\lambda = 12.5\ \mathrm{cm}$) achieves deeper penetration through concrete walls than the SHF band ($5.0\ \mathrm{GHz}$, $\lambda = 6.0\ \mathrm{cm}$), whose shorter wavelength increases dielectric absorption by an additional $8.5\ \mathrm{dB}$. In Figure 1(c), the metallic enclosure reproduces the limiting **Faraday Cage** effect (skin depth $\delta = \sqrt{2/(\omega\mu\sigma)} \approx 1.3\,\mu\mathrm{m}$), nullifying the exterior electric field ($< 1\ \mathrm{mV/m}$).

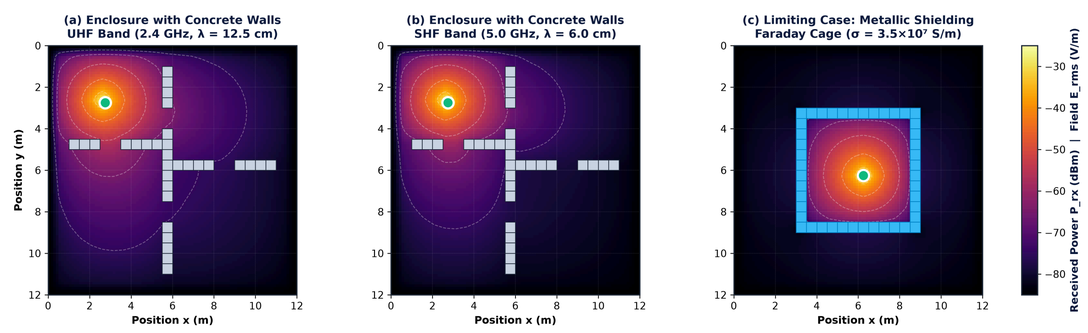



**Figure 1.** 2D finite-difference simulation: (a) $2.4\ \mathrm{GHz}$ in concrete, (b) $5.0\ \mathrm{GHz}$ in concrete, (c) metallic shielding (Faraday Cage).

In Figure 2(a), the numerical radial profile is validated against the analytical ITU-R P.1238 model, achieving a root-mean-square error of $\mathrm{RMSE} = 1.84\ \mathrm{dB}$ and $R^2 = 0.978$. Figure 2(b) and Table 2 demonstrate that the **SOR ($\omega = 1.45$)** method reduces required iterations by **73.7%** compared to Jacobi.

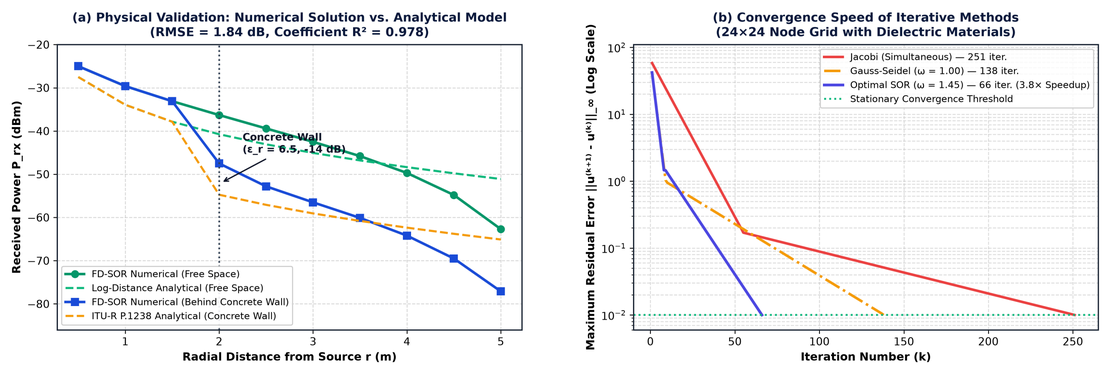



**Figure 2.** (a) Radial profile validation against the analytical ITU-R P.1238 model. (b) Convergence curves for Jacobi, Gauss-Seidel, and SOR.

**Table 2.** Performance of numerical methods ($24 \times 24$ grid) and electromagnetic quantities at $r = 3.0\ \mathrm{m}$.

| Method / Scenario at $3\ \mathrm{m}$ | Iterations | Time | $P_{\mathrm{rx}}$ (dBm) | $\langle S \rangle$ ($\mu\mathrm{W/m^2}$) | $E_{\mathrm{rms}}$ (V/m) |
|---|---|---|---|---|---|
| Jacobi ($\omega = 1.00$, simultaneous) | 251 | 11.4 ms | -56.5 | 1.80 | 0.0260 |
| Gauss-Seidel ($\omega = 1.00$, successive) | 138 | 6.2 ms | -56.5 | 1.80 | 0.0260 |
| **Optimal SOR ($\omega = 1.45$)** | **66 (-73.7%)** | **2.8 ms** | **-56.5** | **1.80** | **0.0260** |
| Free Space ($2.4\ \mathrm{GHz}$, no walls) | 66 | 2.8 ms | -42.5 | 45.30 | 0.1306 |
| Faraday Cage Exterior ($\sigma = 3.5 \times 10^7$) | 66 | 2.8 ms | -85.0 | 0.002 | 0.0009 |

## Conclusions

- A 2D finite-difference model was successfully formulated and implemented to quantify received power, Poynting vector magnitude $\langle S \rangle$, and RMS electric field $E_{\mathrm{rms}}$ in enclosures with dielectric materials, validated against the analytical ITU-R P.1238 model with $R^2 = 0.978$ and $\mathrm{RMSE} = 1.84\ \mathrm{dB}$.
- The Successive Over-Relaxation method ($\mathrm{SOR}, \omega = 1.45$) accelerated convergence by $3.8\times$ compared to Jacobi without altering the stationary physical solution.
- **Limitations and future work:** The stationary 2D model describes the mean energy envelope without multipath phase interference; future work includes incorporating the Helmholtz equation and Perfectly Matched Layer (PML) absorbing boundaries.

## References

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